

Project Design Phase-II Data Flow Diagram & User Stories

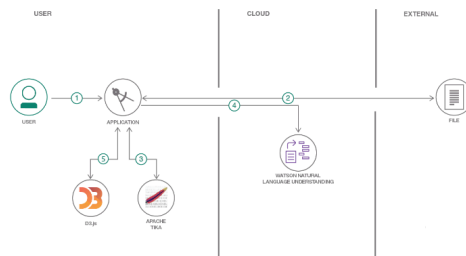
Date	18 July 2026
Team ID	SWTID-2026-2426
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

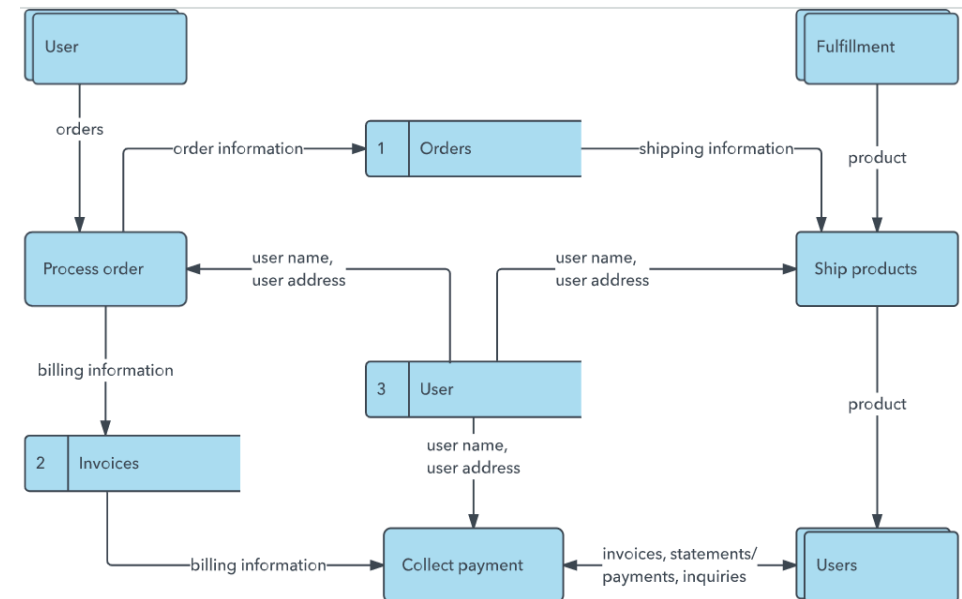
Example: [\(Simplified\)](#)

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Business Analyst	Order & Revenue Trend Analysis	USN-1	As a Business Analyst, I can view total order volume and revenue trends over time so that I can identify demand patterns.	Dashboard displays trend line by day / week / month	High	Sprint-1
		USN-2	As a Business Analyst, I can filter order data by cuisine and meal type so that I can compare performance across categories.	Filters update all visuals correctly	High	Sprint-1
	City-wise Demand Analysis	USN-3	As a Business Analyst, I can view a city-wise breakdown of order volume so that I can identify top-performing cities.	Map / bar chart ranks cities by order volume	High	Sprint-1
		USN-4	As a Business Analyst, I can drill down into a specific city to see cuisine-wise demand so that I can support local strategy decisions.	Drill-down view loads correct city data	Medium	Sprint-2
	Customer Spending Pattern Analysis	USN-5	As a Business Analyst, I can analyse average order value by meal type so that I can identify high-spending segments.	Average order value is calculated and displayed correctly	Medium	Sprint-2
Operations Manager	Delivery Performance Monitoring	USN-6	As an Operations Manager, I can view delivery time distribution across cities so that I can identify delays.	Chart shows delivery time spread by city	High	Sprint-1
		USN-7	As an Operations Manager, I can compare delivery fees against delivery time so that I can evaluate cost-efficiency.	Scatter / comparison view loads both metrics	Medium	Sprint-2
	Peak Hour & Workload Analysis	USN-8	As an Operations Manager, I can view order volume by hour / mealtime so that I can plan staffing during peak hours.	Chart breaks down orders by hour and mealtime	High	Sprint-1
		USN-9	As an Operations Manager, I can identify cities with high delivery delays so that I can reallocate delivery resources.	Cities are ranked / flagged by delay levels	Medium	Sprint-2
Customer	Cuisine & Meal Preference Insights	USN-10	As a Customer insights viewer, I can see the most preferred cuisines by city so that promotions can be tailored.	Cuisine preference chart filters correctly by city	Medium	Sprint-2
		USN-11	As a Customer insights viewer, I can see peak ordering times (lunch / dinner) so that offers can be timed accordingly.	Chart highlights peak mealtime slots	Low	Sprint-3
All Users	Dashboard & Visualization	USN-12	As a dashboard user, I can interact with filters (city, cuisine, date range) so that I can explore data relevant to my role.	All filters update visuals without errors	High	Sprint-1
		USN-13	As a dashboard user, I can export a summary report so that I can share insights with stakeholders.	Report exports in PDF / image format	Low	Sprint-3
Business Analyst	Data Preparation	USN-14	As a Business Analyst, I can access cleaned and pre-processed order data so that analysis is accurate and reliable.	Data has no duplicates / nulls before loading into Tableau	High	Sprint-1